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The Logistics Performance Effect in International Trade

Azmat GANI[a]

[a] *Associate Professor, Sultan Qaboos University, Oman, E-mail:azmat@squ.edu.om*

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ABSTRACT

The continuous growth in world trade depends on the efficiency of trade support structures such as the logistics services. Despite logistics integral role in supporting commercial activities, there has generally been a low level of analysis and trade policy research focus from trade practitioners. This paper explores the effect of logistics performance in international trade. The analysis draws on overall logistics performance as well as disaggregated measures of logistics specificities data for a large sample of countries. The empirical analysis involved the estimation of standard export and import equations incorporating measures of logistics performance. The findings show that the overall logistics performance is positively and statistically significantly correlated with exports and imports. The analysis is also extended by investigating if logistics specificities mattered for international trade. The findings reveal that several dimensions capturing logistics performance have statistically significant and positive effect, mostly on exports. The main policy implication is that continuous investment in logistics infrastructure and services can positively impact international trade.

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1. Introduction

Transport and logistics services facilitate international trade and play an important role in the growth and development of the local economy. The quality and efficiency of logistics services can matter for international trade as a weak logistics infrastructure and operational processes can be a major obstacle to global trade integration (Devlin and Yee, 2005). On the contrary, an improved trade related logistics, combined with a liberalized economic environment, can increase trade volume and economies of scale and scope in distribution and production activities (Lakshman et al. 2001).

Logistics services provide sectoral connections within the local economy. It also connects the domestic economy to the international economy. The connectivity of various inter-dependent production sectors (agriculture, manufacturing, agri-food, tourism, amongst others) of the domestic economy is strengthened through an efficient transport and logistics systems as one of the motives of producers is to securely transport their goods to consumers in a cost-effective way with minimal time lags. Song and Yeo (2017) provided a comprehensive analysis of air transport network of 1060 airports in 173 countries. This is one important study that unfolded an in-depth understanding of the connectivity in the

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air transport sector. Similarly, the domestic producer's ability to participate and reap the benefits of the growing global value chains, GVCs, (factories locating various stages of production processes in most efficient locations and combined with integrated use of technologies (containerization, air freight, telecommunications and informatics)) depends on efficient and reliable transport and logistics systems that can support their international linkages. A recent study by Banga (2014) revealed that 67 percent of global value created under GVC accrued to the Organisation for Economic Co-operation and Development (OECD) countries. Banga (2014) also noted that the share of Newly Industrialised Countries (NICs) and five major emerging economies (Brazil, Russia, India, China and South Africa (BRICs)) were 25 and 8 percent respectively. Thus, a better connectivity can lead to direct development outcomes in terms of facilitating the achievement of vital social development goals such as the creation of employment; the distribution of food and medicines (Pasadilla and Shepherd, 2012 and OECD/World Trade Organisation (WTO), 2013); and improving incomes (Porto et al., 2011).

Although, logistics contribution to the national output in a country may not be as competitive as other sectors, the role that logistics plays in supporting the activities within an economy cannot be undermined or overlooked. One well-known connection between transport and logistics and national development is the facilitation of international trade, which, under appropriate circumstances, delivers several other beneficial economic and social outcomes (OECD/WTO, 2013). The transport and logistic sector is an integral part in terms of facilitating international trade as it allows firms to effectively complete imports and exports of goods and services and associated transactions.

The continuing rise of world trade and the desire by many countries to speed up the pace of integration within the global trading system will depend not only on maintaining an open global economic system but improving the quantity and efficiency of the support structures such as the logistics services. Poor logistics services such as limited co-ordination among countries on border procedures; inefficiency of customs clearance process at the ports; fragmented and poor quality of transportation related infrastructure; costly and infrequent shipping (with long and indirect shipping routes); delays in tracking and tracing consignments; delays in terminal handling and clearance of goods; absence of cool storage facilities at ports; and the inability to certify product quality; amongst others; can cause significant hindrance to international trade.

The forces of trade liberalisation will continue to drive countries around the world to achieve greater participation in, and reap the benefits of the globalising world that offers growing market opportunities. It is the level of development in the domestic as well as international logistics services that can be a critical element in terms of allowing countries to trade without many constraints and at lower costs. While improved overall logistics services can be an important step towards shaping long-term trade facilitation, whether the level of logistics services facilitates more trade is an important empirical question. This issue deserves further investigation as empirical studies from this perspective are rare.

Despite logistics integral role in supporting commercial activities, there is generally a low level of analysis as well as trade policy research focus from an applied economics perspective. Available literature points to the fact that there is little detailed empirical work on the effects of logistics on trade performance. One likely cause of this deficiency is the absence of numerical measures capturing logistics performance on a consistent and timely basis with large country coverage and sector specificity (Shepherd, 2011). While trade researchers have provided some qualitative assessment of the developments in logistics services at country level, an empirical investigation of the impact of logistics on trade could also provide an additional useful guide to trade policy makers as well as logistics operators.

The purpose of this paper is to empirically examine the effect of logistics on international trade. Using regression analysis, the analytical framework in this study uses logistics data from sixty countries, pooled for years 2007, 2010, 2012, and 2014. It estimates an export and an import equation. The analysis is also extended by investigating if the specific dimensions of logistics performance (logistics specificities) matter for traders. In doing so, the paper while complementing past studies on logistics-trade relationship, sheds some new light on the importance of logistics achievements in facilitating international trade. The rest of the paper is structured as follows. Section two reviews the literature. Section three presents the analytical framework. Section four presents the findings. Section five concludes and discusses the policy implications.

2. Literature Review

2.1. *Logistics Performance: An Overview*

Globally, the size of the logistics sector is not clearly known. Shepherd's (2011) analysis of logistics data, covering 45 countries, revealed that, on average, the logistics sector accounts for about 5 percent of the gross domestic product (GDP), with a range of 2 percent to 12 percent. Given the pace at which the world trade has been increasing since 2000 (noted below), the contribution of the logistics sector to the national output in many countries is likely to accelerate as the pace of trade liberalisation strengthens and countries become more and more outward oriented.

Data constraints have made research on logistics formidable. Recent initiatives by the World Bank in terms of compiling the Logistics Performance Index (LPI) for several countries across the globe have made it possible to get a reasonable understanding of achievements in logistics at the national level for several countries around the world. The World Bank has been producing some measures of the LPI annually since 2007. To-date, published data measuring logistic performance in the World Bank member countries are available for years 2007, 2010, 2012, 2014 and 2016. The World Bank's Overall Logistics Performance Index is measured on a scale of 1 (low) to 5 (high). This measure is the weighted average of the country scores covering six sub-dimensions of logistics performance. These sub-dimensions include: the ability to track and trace consignments; the competence and quality of logistics services; the ease of arranging competitively priced shipments; the efficiency of customs clearance process; the frequency with which shipments reach consignee within scheduled or expected time; and the quality of trade and transport-related infrastructure. All these sub-dimensions are also measured on a scale of 1 (low) to 5 (high). The LPI measures on-the-ground trade logistics performance, helping national leaders, key policymakers, and private sector traders understand the challenges they and their trading partners face in reducing logistical barriers to international commerce.

The quality of logistics performance can be gauged from the World Bank data to give a modest indication of logistics performance across countries around the world at different levels of development. Figure 1 depicts the average overall logistics performance index by income category of countries around the world. According to Figure 1, logistics achievements differ markedly across income category of countries. There

are significant gaps in logistics performance between the high-income group countries and the low and middle-income group of countries. Arvis et al. (2012) with regard to the "logistics gap" between the high-income countries argued that the transport and logistics GVCs in the developing countries, including the support for integration into other GVCs, still face very significant obstacles.

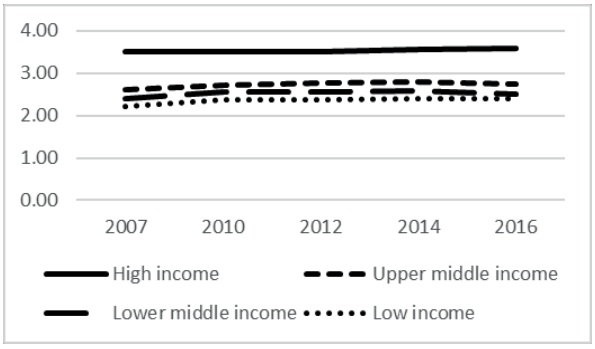


Fig. 1. Trends in overall logistics performance index (1=low to 5=high)

While there are gaps in the overall LPI between the high-income economies and the low and middle-income economies, this has worsened slightly in the lower and upper-income economies since 2007. At the same time, the volume of world trade has also increased more significantly than the rate of improvement in logistical infrastructure and services. For example, according to the WTO (2014a, Tables 1.5 and 1.6), world merchandise exports increased from US$7,380 billion in 2003 to US$18,301 billion in 2013, while world merchandise imports increased from US$7,696 billion to US$18,409 billion over the same period. Part of this trend is explained by the rise of the developing economies trade as a result of their shift towards outward looking economic policies as well as the advancements in new transport and communication technologies. For example, the share of the developing countries world output increased from 23 percent to 40 percent between 2000 and 2012 respectively in purchasing power parity terms and during the same period their share of world trade also rose from 33 percent to 48 percent respectively (WTO, 2014b, p.42). This pattern is likely to continue as a result of the expansion of the GVCs.

2.2. *Studies on Logistics and Trade Links*

As noted in the discussion above, the global economic environment has seen a significant increase in the volume of goods traded over the last two decades. Naturally the volume of goods traded between two countries is largely influenced by the availability and the quality of logistics services, among other things, both in the exporter and importer countries. While the study of the effect of logistics on trade is of much importance in the business context, according to Rutner, Aviles and Cox (2012), historical evidence suggests that the earliest works concerning logistics and its widespread use originated in the military (the method used to implement strategy and tactics) and following the Second World War, logistics has been characterized into two broad fields – business and military (reviewed in Kent and Flint, 1997) with a growing emphasis on the civilian logistics or the business logistics. It is this business logistics aspect that is of main focus in this study.

Logistics in the commercial business context and in particular, trade logistics, include the range of services and processes that are involved in moving goods safely and securely from one country to another. Goods crossing national borders go through a range of processes that include, but are not limited to, satisfying customs and administrative procedures; effective organization and management of international shipment operations; timely tracking and tracing of shipments; and ensuring high quality of transport and information technology infrastructures (Korinek and Sourdin, 2011). Much of the operational functions of a range of these services in many countries are performed mostly by the private sector, with regulatory and other forms of institutional support from the government. The abovementioned logistics functional areas are also linked to each other and so this inter-linked chain of processes involved in moving inputs as well as the finished goods from the producers to consumers within domestic economies as well as across different borders is integral in facilitating international trade.

Although there is a body of literature examining the links between logistics and trade facilitation, there is relatively little analytical work dealing specifically with the effects of logistical services on trade. Existing studies have emphasized the effects of logistics on economic growth or trade in a broader context. The study of logistics, in particular, the specificities or the dimensions that determine logistics and their effects on trade is largely absent and certainly deserves more attention. As noted in the previous section, much of the shortfall in such studies is due to the absence of published logistics data to quantify the effects of logistics services on trade.

Existing studies on logistics and trade generally tend to show consistent outcomes and point to a general direction that there is a positive correlation between improved logistics and more trade. For example, Arvis et al. (2007 and 2010) presented descriptive statistics suggesting a positive association between logistics performance and important outcome indicators, such as trade openness. Dee and Findlay (2006) in their survey of literature on trade facilitation indicated that infrastructure quality is associated with significant increases in trade. Dollar et al. (2004) study revealed that firms in countries subjected to better investment climate with better logistics, had a higher probability in increasing their exports. Saslavsky and Shepherd (2012) confirmed that logistics performance mattered more for trade with GVCs than for other types of trade. Hoekman and Nicita's (2010) analysis of LPI in a gravity model of trade revealed a significant positive association between logistics performance and trade intensity. These authors also noted that the effect was quantitatively important: increasing the average low-income country's LPI score to the middle-income average would increase trade by around 15 percent. Yip (2012) also incorporated gravity modelling framework to analyse grain trade between developed and developing countries, revealing distance (representing transport costs) as a significant friction. In their analysis, Hausman, Lee and Subramanian (2012) argued that logistics performance between two countries can be a significant determinant of total landed cost. Recent studies such as OECD (2013) and Portugal-Perez and Wilson (2012) found that trade facilitation measures not only increased imports but also boosted exports through better access to inputs for production and greater participation in global and regional value chains.

2.3. *Transportation and Logistics*

Transport and logistics service (understood as air and maritime connectivity and logistics performance) is wide and varied. It includes the whole range of transportation infrastructure (high speed roads, terminal facilities, cool storage warehouses, and runaway length) and services from seaports, land based transportation systems as well as air transport, the

corridors within (roads, railways, sea-lanes) and the interconnectedness with these modes of transport. Transport is the single most expensive component of trade logistics and an adequate infrastructure is required to facilitate transportation (OECD, 2011a). It is difficult for a manufacturer to export at a competitive price or import at a competitive cost if the transport and logistics sector is inefficient or even dysfunctional. High prices, poor service, and a lack of certainty in transport and logistics can translate into the effective isolation of a country from world markets. The rise of the GVCs is increasingly becoming significant as the participation in GVCs in manufacturing is one part of the economy that depends fundamentally on transport and logistics for its success (WTO/OECD, 2013). It is the efficiency in the transport and logistics that can affect trade costs as Arvis et al. (2012) argued that transport and logistics contribute as much to the costs of international trade as geographical distance.

Empirical studies provide evidence that inefficiencies within the transportation infrastructure can have adverse effects on trade. For example, Dettmer, Fretag and Draper (2014) in their study on air cargo in South Africa concluded that a more liberal market for air cargo services could reduce transport costs and promote further integration. While ports are found to have biggest impact on trade (Nordas and Pier martini, 2004), port efficiency can impose a significant effect on transport costs. A study by the OECD (2011) revealed that a one day decrease in time spent at sea could increase trade by about 4.5 percent. Shepherd (2011) found that pre-shipment delays were associated with a lower degree of export diversification in developing countries. According to the OECD/WTO (2013) private sector survey data (Table 3), infrastructure remained a serious constraint in many developing countries. Ando and Kimura (2013) in their study on production linkage of Asia and Europe found that a reduction in services link costs could strengthen competence in production networks in East Asia.

Some previous studies noted that logistics specificities can also affect trade. Nordas et al. (2006) in their study found that longer times taken to process exports and imports were associated with reduced trade volumes. Djankov et al. (2006) focused specifically on the time it took to move containerized products from the manufacturing site to a ship in the nearest port. Their findings revealed that each additional day that a product was delayed while awaiting shipment, trade was reduced by more than 1 percent. Hummel (2001) examined the impact of time on trade flows using data for US imports. His study revealed that a reduction in international shipping times by one day was associated with an increase in trade by 0.8 percent. Djankov et al. (2010) in their study on the time taken to export showed that an additional day's delay in an average country was equivalent to moving it away from its trading partners by about 70 kilometres. Martincus et al. (2013) using export data for Uruguay found that customs delays on their own were responsible for significant trade reductions.

2.4. *Governing Trade Logistics*

The governance of logistics can matter for improved trade. Public policies relating to trade logistics are crucial as the efficiency of logistics depends on well-designed government policies. Effective trade logistics is also related to the institutional aspects of logistics such as government regulations, firm level administrative and operational procedures, supply chains and national trade procedures for inward and outward movement of goods. The rapid expansion in international trade volumes as noticed over the last two decades also demands streamlining of operational bureaucracy such as cumbersome, costly and time-consuming trade procedures. Complicated trade administrative procedures can create an environment for corruption to arise that can create further inefficiencies. Malik and Awadallah (2013) have argued that these behind the border barriers create costly frictions in transport, communication and services. These authors also argued that the movement of goods and service governed by restrictions often deny any economic logic in the case of the Middle East and North African countries.

In countries with poor governance and where institutions governing trade transactions and procedures are weak, traders can face significant difficulties in dealing with public sector employees such as customs officials. They can complicate customs clearance through time consuming inspection procedures (requiring unnecessary signatures for clearing goods); customs official's absenteeism from work; and poor lines of communication and unnecessary coordination with many government agencies. Further complications arise when bribery is demanded by poorly remunerated customs workers. For example, Shepherd's (2010) study based on firm-level data showed that poorer trade facilitation (measured by longer lead times to export and import) was associated with higher reported levels of trade-related corruption, as poor performance gave firms an incentive to flout the rules by paying "speed money." Similar findings were reached in an earlier study by Hummel (2001) who found that shippers were willing to pay a premium for a faster delivery. The OECD/WTO (2013) noted that transport and logistics operators were unfortunately in a perfect position to be made subject to a "hold up" problem in the face of (potentially) corrupt police, customs agents or other officials. In the low and middle-income countries, workers in border control areas are generally poorly paid and public servants have an incentive to top up their incomes by asking bribes. Thus, transport and logistics operators are confronted with binding constraints such as access to crucial facilities (roads, ports) or completion of necessary processes (customs formalities and other border checks). In order to overcome such binding constraints, logistic operators may be unnecessarily drawn into unofficial payments or payments of "speed money" (Olken and Barron, 2009).

Market structure can also complicate logistic services and could act as a barrier towards improving operational efficiencies. In the absence of a competitive logistics services (or the presence of imperfect markets), can translate into higher costs for the traders as competition in providing logical services can be obstructed due to powerful monopolies or duopolies. Devlin and Yee (2005) noted that, the trucking industry, a key conduit for goods in most Middle East and North African countries, had a general prevalence of freight monopolies and pricing practices were not economically efficient since the high rates levied on non-regulated commodities were used to offset the low rates received for essential commodities. Such practices do prevent the rise of new competitive firms and existing firms are unable to exploit the industry economies of scale. Devlin and Yee (2005) further noted that airfreight in many Middle East and North African countries are largely provided by state-owned airlines and airports, with competition constrained by licensing regimes and state regulating all the commercial port services. Encouraging entry by private operators and eliminating restrictive licensing regimes are important steps towards improved efficiency and quality of services in the logistics sector.

3. Methodology

3.1. *Conceptual Framework, Hypotheses and Measures*

The conceptual framework of this study follows the well-established international trade theory. Almost two centuries ago, the theory of comparative advantage based on David Riccardo's work (Riccardo, 1870) postulated that countries can engage in mutually beneficial trade with each other and specialize their production in goods for which their opportunity cost of production (relative to the other country) is lowest. This theory has been generalised and extended from the initial two-good setting to a multiproduct setting (for example, Dornbusch, Fischer and Samuelson, 1977) and multi-country contexts (for example, Eaton and Kortam, 2002).

Building on the theory of comparative advantage, Swedish economist, Eli Heckscher and Bertil Ohlin (his student) provided an alternative model to the Riccardo's (1870) model, widely known as the Heckscher-Ohlin (H-O) model where relative autarky price differences arise from differing factor endowments in countries and differing intensities with which sectors use these factors of production. In essence, the H-O theorem postulated that a country will export the good that uses relatively intensively the factor of production which is relatively abundant in a country.

Given the increased levels of trade of the 20th century between countries with similar characteristics which made it somewhat difficult to explain by comparative advantage, Paul Robin Krugman explained the patterns of international trade and the locality of economic activity in different countries which led to the development of new trade theory (Krugman, 1979). The Krugman (1979) model has two core elements. The first is that the production sector in each country is characterised by monopolistic firms with economies of scale. The second is that utility increases as more variation of a product is available for consumption. The recent advances in trade theory is the focus on micro foundations and effects of trade, that is, the understanding of agents who make decision to engage in international trade. In a general equilibrium framework, Melitz (2003) introduces the extension of the new trade theory. Yip (2012) provides a good review of the importance of new trade theory in analysing bilateral trade. Blonigen and Wilson (2013) provided a comprehensive review of trade theories, including, the comparative advantage theory, the new trade theory and recent advances in trade theory. Past literature has extensively addressed the long-term influence of trade on economic growth and development of nations, for example, Greenaway, Morgan and Wright (2002); Yanikkaya (2003); Santos-Paulino (2005); Kneller, Morgan and Kanchanahatakij (2008); Chang, Kattani and Loayza (2009); and Gani and Scrimgeour (2016).

Although the mainstream trade theory and empirical literatures reviewed above, and the previous section has addressed several issues relating to trade and its connectedness to nations economic growth and development; studies empirically examining the direct effect of logistics on international trade is still rare. Much of the studies connecting logistics to trade are industry based as reviewed in section 2.0. This has also constrained the direct linkage of the findings of several of the studies (reviewed in section 2.0) to the conceptual framework as discussed above. The analytical framework and analysis that follows below is expected to make some connection on the importance of logistics with the conceptual framework. In view of the above discussion, the empirical model explaining the effects of logistics on trade is represented by equation (1).

$$Y_{i,t} = \alpha X_{i,t} + \beta_2 Logistics_{i,t} + \varepsilon_{i,t} \quad (1)$$

where, Y is total trade, matrix X includes a set of control variables, and ε is the error term. Variable Y includes exports of goods and services as a percentage of gross domestic product (EX) and imports of goods and services as a percentage of gross domestic product (IM). The control variables in matrix "X" are the rate of change of relative prices (RP); domestic economic growth rate (GDP-D) for the import model; world economic growth rate (GDP-W) for the export model; trade liberalisation (TL); contract enforcement (CE); common language (CL); and common border (CB). The core variable of interest is Logistics and includes an overall measure of logistics performance (LP-O) as well as measures of logistics specificities (discussed in section 3.2). Appendix Table 1 presents a summary of the variable definitions and measures for variables EX, IM, RP, GDP-D, GDP-W, TL, CE, CL and CB. The data sources for variables EX, IM, RP, GDP-D, GDP-W, and CE is World Development Indicators (The World Bank, 2017). TL, CL and CB are captured by dummy variables. In what follows next is a brief theoretical justification of the inclusion of the right-hand-side variables in equation (1).

RP is included as a measure of a country's international price competitiveness as it is regarded as a key determinant of a country's international trade in goods and services (Pugel, 2004). Changes in exchange rates can alter a country's trade pattern as a shift in tastes and preferences toward foreign goods, which leads to an increase in the domestic price of the foreign currency, will make domestic products cheaper when measured in that foreign currency, thereby increasing the competitiveness of a country in terms of exports (Appleyard and Field, 2014, p.49). A recent study by Kim (2017) revealed that exchange rate volatility had a statistically significant negative influence on Korea's seaborne import volume.

GDP-D is included as a control variable as it can matter significantly for importers as a growing domestic economy indicates that the domestic economic environment is suitable for investment, production and sales and is likely to encourage imports. An economy in which domestic income growth remains stagnant may discourage imports.

On the export side, variable GDP-W is included as exporters are likely to positively respond to a growing economy of their trading partners and hence respond to the expected increases in demand for goods and services by increasing their exportable items. Growth in trading partner can also reflect increases in incomes. In a recent study, Ha, Chung and Seo (2016) found that the elasticity of income exerted a significant positive influence on the volume of Korea's trade from ASEAN-5 countries. In another study on international grain trade, Yip (2012) found that importer's GDP led to grain trade growing much faster than exporter's GDP.

The analysis here also controls for trade liberalization, TL, as outward oriented trade policies have been more successful than inward oriented trade policies (Edwards, 1993) while free trade theory also postulates that increases in trade restrictions can have regressive effects on an economy.

CE reflects the extent of compliance. This can impact trade as a lack of enforcement mechanism can discourage economic activity (Dabla-Norris and Freeman, 2004). Poor contract enforcement by the state's legal system raises the costs of trade and therefore lowers the volume of trade (Anderson and Marcouiller, 2002 and Ranjan and Lee, 2007). On the contrary, sound contract enforcement and a good quality legal system enhances the export activities of firms (Zhang, Wang and Qu, 2012).

CL is included as a control variable and has been empirically modelled as a non-tariff barrier to trade (for example, Helliwell, 1999 and Egger and Lassmann, 2012). It is hypothesized that countries sharing a common language with trading partners can stimulate interaction in general and increase trade transactions among the trading partners.

In terms of common border, CB, Hess and van Wincoop (2000) noted that countries within regions are clearly more integrated as they trade more with each other. The hypothesis for CB is that in countries sharing common borders, crossing borders is relatively easy and products across the border may be significantly cheaper and so this can have a positive effect on trade.

3.2. *Sampling and Data Collection*

The source of the logistics data is the World Development Indicators (The World Bank, 2017) and the core measure is the Overall Logistics Performance Index (LP-O), measured on a scale of 1 (low) to 5 (high). This measure is the weighted average of the country scores covering six sub-dimensions of logistics performance. These sub-dimensions include: ability to track and trace consignments (LP-C); competence and quality of logistics services (LP-S); ease of arranging competitively priced shipments (LP-ST); efficiency of customs clearance process (LP-CC); frequency with which shipments reach consignee within scheduled or expected time (LP-F); and quality of trade and transport-related infrastructure (LP-I). All these indicators are measured on a scale of 1 (low) to 5 (high).

The sample includes sixty countries: Australia; Austria; Belgium; Canada; Czech Republic; Denmark; Finland; France; Germany; Greece; Ireland; Italy; Japan; Luxembourg; Netherlands; New Zealand; Norway; Portugal; Slovak Republic; Spain; Sweden; Switzerland; United Kingdom; United States; Algeria; Armenia; Bahrain; Bolivia; Bulgaria; Cameroon; Chile; China; Colombia; Costa Rica; Cote d'Ivoire; Croatia; Dominican Republic; Ecuador; Gabon; The Gambia, Ghana; Hungary; Macedonia; Malaysia; Mexico; Moldova; Nigeria; Pakistan; Paraguay; Philippines; Poland; Romania; Russian Federation; Saudi Arabia; Singapore; Solomon Islands; South Africa; Tunisia; Uruguay; and Venezuela. It should be noted that these countries are chosen because of availability of data on all the variables as identified in equation (1) across four-time periods (2007, 2010, 2012 and 2014). It should be further noted that several countries had data available for the matrix "X" (control variables) but had missing data for logistics measures. As such, missing data on logistics measures meant that several countries had to be dropped from the estimation phase.

4. Analysis and Findings

The empirical analysis adopts the cross-sectional estimation of the data. The data is pooled, and the estimation phase combines cross-country and time-series data involving sixty countries and four-time periods (2007, 2010, 2012 and 2014). These four years are chosen due to availability of published data on logistics performance as mentioned above. In addition, data on other control variables had to be consistent for these four years for all the countries in the sample. Published data on logistics performance index are not available for years 2008, 2009, 2011 and 2013. These years with missing data are controlled through a time dummy variable in the estimation process (variable TD in Tables 1, 2 and 3). In the estimation phase, I also control for the effects of the low and middle-income countries (variable LMC in Tables 1, 2 and 3) as these income group of countries have lower levels of logistic achievements as opposed to high-income countries. LMC is a dummy variable taking a value of 1 if a country is low or middle-income and 0 otherwise.

The descriptive statistics are presented in Appendix Table 2. Table 1 reports the results of the effects of LP-O on exports and imports. The fit of the models is as expected with lower values given the use of the pooled data. The parameter estimates obtained are corrected for cross-sectional hetroskadasticity. A number of variables produced statistically significant effect. In Tables 1, 2 and 3; the asterisks, *; **; and ***; indicate that the coefficient is statistically significant at the 1, 5, and 10 percent levels, respectively. The *t*-statistics are in parentheses while (---) indicates that the corresponding variable was not estimated. In all the Tables below, N indicates the number of observations and rho is the cross-section rho.

Table 1

The effect of achievements in overall logistics (LP-O) on trade

Variables	Expected effects	Log-linear export model	Log-linear import model
CONSTANT	(+)	3.463 (4.887)*	3.398 (5.786)*
EX_{t-1}	(+)	0.044 (0.764)	…
IM_{t-1}	(+/-)	…	0.109 (1.955)**
RP	(-)	-0.0003 (0.738)	-0.002 (0.556)
GDP-W	(+)	-0.011 (0.290)	…
GDP-D	(+)	…	0.010 (0.903)
TL	(+)	-0.239 (1.075)	-0.292 (1.409)
LMC	(-)	-0.32341 (2.964)*	-0.174 (1.735)***
CE	(-)	-0.0003 (1.934)***	-0.0003 (2.330)**
CL	(+)	-0.083 (1.036)	-0.143 (1.957)**
CB	(+)	0.217 (2.767)**	0.274 (3.805) *
LP-O	(+)	0.356 (3.061)*	0.148 (1.658)***
TD	(+/-)	-0.244 (0.564)	0.072 (0.188)
N		240	240
F-statistics		4.189	4.09
Adjusted R^2		0.15	0.15
Rho		-0.092	-0.088

The core variable of interest is LP-O. The result in Table (1) shows that the sign on coefficient LP-O is as expected, positive and statistically significant at the 1 percent level for the export equation. In terms of imports, the result in Table (1) shows that that the coefficient LP-O has the expected positive sign and statistically significant at the 10 percent level. The statistically positive correlation of LP-O with exports and imports as shown in Table 1 leads to the conclusion that the logistics is an important determinant of international trade as improved trade related logistics can facilitate more trade. This is an important empirical finding in terms of trade facilitation. This finding is consistent with similar arguments raised in earlier studies such as Devlin and Yee (2005) and Lakshman et al. (2001).

In estimating equation (1), many standard variables hypothesised to determine trade, other than then LP-O, were also controlled for the export

and import equations as represented in equation (1). The results in Table (1) of the control variables indicate that in terms of exports, CE and CB are other core determinants. On the import side, lagged imports, CE and CB are found to be significant determinants.

Given the positive effect of LP-O on exports and imports, the empirical analysis was extended further to test if some of the specific aspects of logistics (logistics specificities) mattered for trade. The effects of six disaggregated measures of logistics were tested for exports and imports. These were: the ability to track and trace consignments (LP-C); competence and quality of logistics services (LP-S); ease of arranging competitively priced shipments (LP-ST); efficiency of customs clearance process (LP-CC); frequency with which shipments reach consignee within scheduled or expected time (LP-F); and quality of trade and transport-related infrastructure (LP-I). These six measures of logistics specificities were tested in several specifications one at a time to avoid the multicollinearity effect (the variable overlapping effect). Table 2 presents the regression results of the effect of logistic specificities on exports.

Table 2

The effect of logistics specificities on exports

	Specifications					
Variable	1	2	3	4	5	6
CONSTANT	3.974 5.346)*	3.735 (5.261)*	3.382 (4.6155)*	3.369 (5.387)*	3.402 (4.764)*	3.723 (5.381)*
EX_{t-1}	0.048 (0.826)	0.030 (0.648)	0.042 (0.729)	0.044 (0.784)	0.049 (0.850)	0.042 (0.750)
RP	-0.003 (0.698	-0.003 (0.633)	-0.002 (0.472)	-0.004 (0.929)	-0.003 (0.693)	-0.003 (0.719)
GDP-W	-0.021 (0.559)	-0.013 (0.333)	-0.009 (0.239)	-0.007 (0.179)	-0.025 (0.681)	-0.009 (0.261
TL	-0.214 (0.952)	-0.206 (0.921)	-0.212 (0.961)	-0.222 (1.001)	-0.266 (1.185)	-0.194 (0.869)
LMC	0.219 (2.100)**	-0.253 (2.312)**	-0.227 (2.416)**	-0.361 (3.413)*	-0.275 (2.737)**	-0.291 (2.679)*
CE	-0.0004 (2373)**	-0.0004 (2.270)*	-0.0003 (2.188)**	-0.0003 (1.858)***	-0.0004 (2.189)**	-0.0003 (1.936)**
CL	-0.099 (1.225)	-0.096 (1.202)	-0.092 (1.152)	-0.094 (1.197)	-0.058 (0.711)	-0.099 (1.239)
CB	0.224 (2.808)**	0.226 (2.847)*	0.231 (2.935)*	0.210 (2.707)*	0.227 (2.890)**	0.219 (2.781**
LP-C	0.227 (2.614)**	…	…	…	…	…
LP-S	…	0.257 (2.793)*	…	…	…	…
LP-ST	…	…	0.335 (3.338)*	…	…	…
LP-CC	…	…	…	0.368 (4.253)*	…	…
LP-F	…	…	…	…	0.323 (3.574) *	…
LP-I	…	…	…	…	…	0.259 (3.277)*
TD	-0.136 (0.427)	0-.235 (0.538)	-0.231 (0.529)	-0.322 (0.767)	-0.246 (0.688)	-0.232 (0.543)
N	240	240	240	240	240	240
F-statistics	3.50	3.64	3.92	4.735	4.099	3.927
Adjusted R^2	0.13	0.14	0.15	0.17	0.15	0.15
Rho	-0.093	-0.099	-0.088	-0.090	-0.086	-0.093

The results in Table (2) revealed positive and statistically significant effect of all six logistics specificities on exports. Variables LMC, CE and CB are also found to be statistically significantly correlated with exports. The negative coefficient of LMC indicates the regressive effect of lower achievements in logistics on exports in the low and middle-income group of countries.

Table 3 presents the regression results of the effect of logistic specificities on imports. According to the results in Table 3, two of the six logistics specificities coefficients, LP-ST and LP-CC, have the expected positive signs and are statistically significant at the 5 percent level. The remaining four logistics specificities (LP-C; LP-S; LP-F; and LP-I) while having a positive coefficient, are statistically insignificant in the import equation. In estimating the effects of logistics specificities on the import equation, all standard variables hypothesised to determine trade were also controlled. The results in Table (3) of the control variables indicate that in terms of imports, lagged imports, CE and CB are statistically significant determinants.

Table 3

The effect of logistics specificities on imports

	Specifications					
Variable	1	2	3	4	5	6
CONSTANT	3.552 (6.064)*	3.549 (6.110)*	3.277 (5.472)*	3.421 (6.069)*	3.344 (5.521)*	3.549 (6.189)*
IMt-1	0.108 (1.942)**	0.107 (1.914)**	0.107 (1.927)**	0.110 (1.996)**	0.109 (1.967)**	0.107 (1.929)**
RP	-0.002 (0.469)	-0.002 (0.474)	-0.002 (0.471)	-0.003 (0.739)	-0.002 (0.487)	-0.002 (0.495)
GDP-D	0.010 (0.925)	0.010 (0.909)	0.011 (0.739)	0.009 (0.912)	0.010 (0.902)	0.011 (0.949)
TL	-0.275 (1.321)	-0.273 (1.318)	-0.288 (1.404)	-0.294 (1.432)	-0.302 (1.447)	-0.270 (1.305)
LMC	-0.109 (1.073)	-0.123 (1.158)	-0.158 (1.714)***	-0.240 (2.291)**	-0.149 (1.504)	-0.133 (1.261)
CE	-0.0004 (2.626)*	-0.0004 (2.577)**	-0.0003 (2.352)**	-0.0003 (2.179)**	-0.0004 (2.470)**	-0.0004 (2.442)**
CL	-0.153 (2.094)**	-0.152 (2.071)**	-0.143 (1.973)**	-0.142 (1.973)**	-0.134 (1.802)***	-0.152 (2.094)**
CB	0.280 (3.864)*	0.280 (3.873)*	0.277 (3.863)*	0.265 (3.707)*	0.279 (3.879)*	0.279 (3.849)*
LP-C	0.071 (0.898)	…	…	…	…	…
LP-S	…	0.083 (1.003)	…	…	…	…
LP-ST	…	…	0.181 (1.993)**	…	…	…
LP-CC	…	…	…	0.202 (2.561)**	…	…
LP-F	…	…	…	…	0.129 (1.543)	…
LP-I	…	…	…	…	…	0.083 (1.152)
TD	0.108 (0.283)	0.087 (0.225)	-0.063 (0.164)	0.008 (0.019)	0.107 (0.279)	0.089 (0.234)
N	240	240	240	240	240	240
F	3.85	3.91	4.17	4.53	4.02	3.93
Adjusted R^2	0.14	0.15	0.15	0.17	0.15	0.15
Rho	-0.87	-0.09	-0.82	-0.087	-0.85	-0.089

5. Conclusion

This paper examined the effects of logistics performance in international trade using cross-country data for a large sample of countries. The findings revealed statistically significant positive correlation of overall logistics performance with exports and imports. The study also extended the analysis by examining six measures of logistics specificities on trade, providing strong confirmation of positive and statistically significant correlation of all the six of the logistics specificities with exports. In terms of imports, two of the six measures of logistics specificities were found to be positively and statistically significantly correlated with imports.

5.1. *Discussion and Implications*

Given the significant rise in world trade over the last two decades as noted in section 2, it is quite likely that countries around the globe will continue the process of trade liberalization with greater strength and magnitude to achieve wider trade integration and reap the benefits of the growing opportunities that the global market has to offer. Thus, the continuing rise of world trade and the desire by many countries to be part of an integrated global trading system will depend not only on maintaining an open global economic system but improving the quantity and efficiency of their logistics services. It is the speed and the quality of investments in logistics services that can be a critical element in terms of facilitating greater volume of trade. The findings of this study lead to the conclusion that the quality of logistics is an important determinant of international trade. An improved logistics will positively be influential in facilitating greater volume of international trade.

The findings of this study provide strong evidence of the positive role that logistics plays in increasing trade. The discussion in this paper also revealed that logistics achievements in the low and middle-income group of countries are at lower levels than the high-income countries. As discussed in the review of the literature in this study, several aspects of logistics specifics to countries and regions of the developing world still pose substantial difficulties in easing up the flow of goods in a cost effective and timely manner. For example, geographical hurdles, cumbersome customs management with excessive and opaque procedural requirements and physical inspection of goods, corrupt border control state officials, absence of coherent national transport strategy, and lack of competition in land, sea and air freight services continue to be major hindrances to trade. It is the public and private sector forging coherent national working partnerships relating to investments in logistics infrastructure and better logistics governance and co-ordination as well as integration of trade related institutions that is likely to help modernize the logistics services in countries with low levels of logistics performance and overcome some of the above mentioned logistical constraints.

5.2. *Limitations, and Future Research*

One of the limitations of this research is to do with the availability of published long-term time-series data for empirical analysis. The current empirical analysis adopts the cross-sectional estimation, where data were pooled and the estimation phase combined cross-country and time-series data involving sixty countries and four time periods (2007, 2010, 2012 and 2014). While this empirical approach increases the number of observations and enriches the analysis, the implications of the findings here are largely applicable to the set of countries considered in this study. Once published long-term time series data on measures of logistic variables considered in this study becomes available, future research can extend this analysis on a country-specific basis to ascertain more precise implications at country-specific level.

Appendix Table 1 – Variable description

Variable	Description	Measure
EX	Exports	Exports of goods and services as a percentage of gross domestic product
IM	Imports	Imports of goods and services as a percentage of gross domestic product
RP	Relative prices	The real exchange rate index (2005=100),
GDP-W	World economic growth	The real GDP growth rate (annual percentage) of each of the sixty countries in the sample
GDP-D	Domestic economic growth	The real GDP growth rate (annual percentage) of the world.
TL	Trade liberalisation	A dummy variable, taking a value of 1 for membership to WTO and 0 otherwise. Country membership to WTO was sourced from WTO website (http://www.wto.org)
CE	Contract enforcement	The number of calendar days from filing of the lawsuit in a court until the final determination and in appropriate cases, payment.
CL	Common language	A dummy variable, taking a value of 1 if a country had common language with its main export partner and 0 otherwise.
CB	Common border	Measured by a dummy variable, taking a value of 1 if a country had common border with its main export partner and 0 otherwise.

Appendix Table 2 –Descriptive statistics

Variable	N	Mean	Standard deviation	Variance	Min	Max
EX	240	47.500	34.423	1185.0	11.498	214.74
IM	240	46.363	29.243	855.15	12.450	183.92
RP	240	99.658	7.7923	60.719	60.585	128.13
GDP-W	240	3.9077	0.87826	0.77134	2.8310	5.0000
GDP-D	240	3.6248	3.1185	9.7248	-6.5721	15.240
CE	240	563.38	235.23	55331.	120.00	1580.0
LP-O	240	3.2026	0.59247	0.35102	2.0600	4.1900
LP-C	240	3.2414	0.63456	0.40266	1.9167	4.2700
LP-S	240	3.1660	0.63606	0.40457	1.7500	4.3200
LP-ST	240	3.1061	0.48888	0.23901	1.6700	4.0500
LPI-CC	240	2.9885	0.65656	0.43107	1.6000	4.2078
LP-F	240	3.6115	0.57295	0.32827	2.3000	4.7059
LP-I	240	3.1138	0.73539	0.54080	1.7800	4.3400

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References

ANDERSON, J. E. and MARCOUILLER, D. (2002). "Insecurity and the pattern of trade: an empirical investigation", *Review of Economics and Statistics*, Vol. 84, No. 2, pp. 342-52.

ANDO, M. and KIMURA, F. (2013). "Production linkage of Asia and Europe via Central and Eastern Europe", *Journal of Economic Integration*. Vol. 28, pp. 204-240.

APPLEYARD D. R, and FIELD A. J JR. (2014). "*International Economics*". (Eighth edition). McGraw Hill Irwin, New York

ARVIS J.-F, MUSTRA M., OJALA. J., SHEPHERD, B., and SASLAVSKY, D. (2012). "*Connecting to Compete: Trade Logistics in the Global Economy*". Washington, D.C.: World Bank.

ARVIS, J.-F., MUSTRA, M., PANZER, J. OJALA. L, and NAULA, T. (2007). "*Connecting to Compete 2007: Trade Logistics in the Global Economy*". Washington, D.C.: The World Bank.

ARVIS, J.-F., MUSTRA, M., OJALA, L., SHEPHERD, B., and SASLAVSKY. D. (2010). "*Connecting to Compete 2010: Trade Logistics in the Global Economy*", Washington, D.C.: The World Bank.

BANGA, R. (2014). "Linking into global value chains is not sufficient: do you export domestic value-added contents"? *Journal of Economic Integration,* Vol. 29, pp. 267-297.

BLONIGEN, B. A. and WILSON, W. W. (2013), "The growth and pattern of international trade," *Maritime Policy and Management,* Vol. 40, No. 7, pp. 618-635.

CHANG, R., KATTANI, L., and LOAYZA, N.V. (2009). "Openness can be good for growth: the role of policy complementarities", *Journal of Economic Development,* Vol. 90, pp. 33-49.

DABLA-NORRIS, E. and FREEMAN, S. (2004). "Enforcement of property rights and underdevelopment", *Review of Development Economics,* Vol.8, No. 3, pp. 391-405

DEE, P., and FINDLAY, C. (2006). "*Trade Facilitation: What, Why, How, Where and When*"? Working paper, Asian Development Bank, Manila.

DETTMER, B., FRETAG, A., and DRAPER, P. (2014.) "Air cargo beyond trade barriers in Africa", *Journal of Economic Integration,* Vol. 29, pp. 95-138.

DEVLIN, J., and YEE, P. (2005). "Trade logistics in developing countries: the case of the Middle East and North Africa", *The World Economy,* Vol. 28, pp. 435-456.

DJANKOV, S., FREUND, C. and PHAM, S. (2006). "*Trading on time*", Working paper No. 3909, The World Bank, Washington, DC.

DJANKOV, D., FREUND, C., and PHAM, C. (2010). "*Trading on time*", *Review of Economics and Statistics,* Vol. 92, pp. 166-173.

DOLLAR, D., HALLWARD-DRIEMEIER, M., and MENGISTAE, T. (2004). "*Investment Climate and International Integration*", Working paper No. 3323, The World Bank, Washington, DC.

DORNBUSCH, R., FISCHER, S., and SAMUELSON, P. A. (1977). "Comparative advantage, trade and payments in a Riccardian model with a continuum of goods," *The American Economic Review,* Vol. 67, No. 5, pp. 823-839.

EATON, J. and KORTUM, S. (2002). "Technology, geography and trade," *Econometrica,* Vol. 70, No.5, pp. 1741-1779.

EGGER, P. H., and LASSMAN, A. (2012). "The language effect in international trade: a meta-analysis", *Economic Letters,* Vol. 116, pp. 221-24.

EDWARDS, S. (1993). "Openness, trade liberalisation and growth in developing countries", *Journal of Economic Literature,* Vol. 31, No. 3, pp. 1358-1393.

GANI, A. and SCRIMGEOUR, F. (2016), "New Zealand's trade with Asia and the role of good governance," *International Review of Economics and Finance,* Vol. 42, pp. 36-53

GREENAWAY, D., MORGAN, W., and WRIGHT, P. (2002). "Trade liberalisation and growth in developing countries", *Journal of Development Economics,* Vol. 67, pp. 229-244.

HA, Y. S., CHUNG, K. J., and SEO, J. S. (2016), "An analysis of Korea-ASEAN trade and its implications for the shipping industry in Korea," *The Asian Journal of Shipping and Logistics,* Vol. 32, No. 2, pp. 63-71.

HAUSMAN, W. H., LEE, H. L, and SUBRAMANIAM, U. (2012). "The impact of logistics performance on trade", *Production and Operations Management,* Vol. 22, pp. 236-252.

HELLIWELL, J. F. (1999). "Language and trade", In Breton, A., *Exploring the Economics of Language*. Canadian Heritage.

HESS, G. D. and VAN WINCOOP, E. (2000). "*Intranational Macroeconomics*", Cambridge University Press.

HOEKMAN, B. and NICITA, A. (2010). "Assessing the Doha round: market access, transactions costs and aid for trade facilitation", Journal of International Trade and Economic Development, Vol. 19, pp. 65-79.

HUMMEL, D. (2001). "Time as a trade barrier", Working Paper. Available on line at: www.krannert.purdue.edu/faculty/hummelsd/research/time3b.pdf.

KENT, J. L. JR and FLINT, D. J. (1997). "Perspectives on the evolution of logistics thought", *Journal of Business Logistics,* Vol. 18, pp. 15-29.

KIM, C. B. (2017). "Does exchange rate volatility affect Korea's seaborne import volume?" *The Asian Journal of Shipping and Logistics,* Vol. 33, No. 1, pp. 43-50.

KNELLER, R., MORGAN, C.W. and KANCHANAHATAKIJ, S. (2008). "Trade liberalization and economic growth", *The World Economy,* Vol. 31, No. 6, pp. 701-719.

KORINEK, J. and SOURDIN, P. (2011). "To what extent are high-quality logistics services trade facilitating"? OECDTrade Policy Papers, No. 108, OECD Publishing. Available online at: http://dx.doi.org/10.1787/5kggdthrj1zn-en.

KRUGMAN, P. R. (1979). "Increasing returns, monopolistic competition and international trade," *Journal of International Economics,* Vol. 9, No. 4, pp. 469-479.

LAKSHMANAN, T. R, SUBRAMANIAN, U., ANDERSON, W. and LEAUTIER, F. (2001). "Integration of Transport and Trade Facilitation," Directions in Development (Washington, DC: World Bank).

MALIK, A. and AWADALLAH, B. (2013). "The economics of Arab spring", *World Development,* Vol. 45, pp. 296-313.

MARTINCUS, C., CARABALLO, J., and GRAZIANO, A. (2013). "Customs as doorkeepers: what are their effects on international trade", Working Paper, available on line at:
www.usitc.gov/research_and_analysis/documents/Customs_as-Doorkeepers-What_Are_Their_Effects_on_International_Trade.pdf.

MELITZ, M. J. (2003). "The impact of trade on intra-industry reallocations and aggregate industry productivity," *Econometrica*, Vol. 71, No. 6, pp. 1695-1725.

NORDAS, H. K., PINALI, E., and GROSSO, M. G. (2006). "Logistics and time as a trade barrier", OECD Trade Policy Working Paper, No. 35, OECD, Paris.

NORDAS, H. K. and PIERMARTINI, R. (2004). "Infrastructure and trade", WTO Staff Working Paper No. ERSD-2004-04. World Trade Organisation. Geneva.

OECD/WTO. (2013). "Aid for trade and value chains in transport and logistics", World Trade Organisation, Geneva.

OECD. (2013). "Trade facilitation indicators: the potential impact of trade facilitation on developing countries trade", Trade Policy Paper 144, OECD.

OECD. (2011). "To what extent are high-quality logistics services trade facilitating"? Trade Policy Working Paper 108, OECD.

OLKEN, B. and BARRON, P. (2009). "The simple economics of extortion: evidence from trucking in Aceh", *Journal of Political Economy*, Vol. 117, pp. 417-452.

PASADILLA, G. and SHEPHERD, B. (2012). "Trade in Services and Human Development: A First Look at the Links", In P. Sauvé, G. Pasadilla, and M. Mikic (eds.). *Service Sector Reforms: Asia-Pacific Perspectives*. Bangkok: United Nations.

PORTO, G., DEPETRIS, C. N., and OLARREAGA, M. (2011). "Supply Chains in Export Agriculture, Competition, and Poverty in Sub-Saharan Africa", Washington, D.C.: World Bank.

PORTUGAL-PEREZ, A. and WILSON, J. S. (2012). "Export performance and trade facilitation reform: hard and soft infrastructure", *World Development*, Vol 40, pp. 1295–1307.

PUGEL, T.A. (2004). "*International Economics"*, Boston: McGraw-Hill Irwin.

RANJAN, P. and LEE, J. Y. (2007). "Contract enforcement and international trade", *Economics and Politics*, Vol. 39, No. 4, pp. 1177-203.

RICCARDO, D. (1817). "*Principles of Political Economy and Taxation*," London: J. Murray.

RUTNER, S. M., AVILES, M. and COX, S. (2012). "Logistics evolution: a comparison of military and commercial logistics", *The International Journal of Logistics Management*, Vol. 23, pp. 96-118.

SANTOS-PAULINO, A. (2005). "Trade liberalization and economic performance: theory and evidence for development of countries", The World Economy, Vol. 28. No. 6, pp. 783.

SASLAVSKY, D. and SHEPHERD, B. (2012). "Facilitating international production networks: the role of trade logistics", Policy Research Working Paper 6224, World Bank.

SHEPHERD, B. (2011). "Logistics costs and competitiveness: measurement and trade policy applications", Transport Research Support Working Paper, World Bank. Available on line
at:
http://siteresources.worldbank.org/INTTRANSPORT/Resources/336291-1239112757744/5997693-1294344242332/Logistics_costs.pdf.

SHEPHERD, B. (2010). "Speed money: time, corruption, and trade", Working Paper, https://sites.google.com/a/developing-trade.com/home/publications/files/ShepherdWPSpeedMoney26Feb10.pdf?attredirects=0.

SONG, M. G. and YEO, G. T. (2017). "Analysis of the air transport network characteristics of major airports," *The Asian Journal of Shipping and Logistics*, Vol.33, No.3, pp. 117-125.

WORLD BANK. (2017). "*World Development Indicators*," Available online at http://www.worldbank.org/data/reports.aspx?source-world-development-indicators.

WORLD TRADE ORGANISATION. (2014a). "*International Trade Statistics 2014*" Geneva. (Available online at: http://www.wto.org/english/res_e/statis_e/its2014_e?its14_toc-e.htm).

WORLD TRADE ORGANISATION. (2014a). "*International Trade Statistics 2014*," World Trade Organisation. Geneva. (Available online at http://www.wto.org/english/res_e/its2014_e/its14_toc_e.htm.

WORLD TRADE ORGANISATION. (2014b). World Trade Report 2014. Trade and Development: Recent Trends and the Role of WTO. World Trade Organisation. Geneva. (Available online at
http://www.wto.org/english/res_e/publications_e/wtr14_forum_e.htm.

YANIKKAYA, H. (2003). "Trade openness and economic growth: a cross-country empirical investigation", *Journal of Development Economics*, Vol. 72, pp. 57-89

YIP, T. L. (2012). "Seaborne trade between developed and developing countries," *The Asian Journal of Shipping and Logistics*, Vol. 28, No.3, pp. 369-390.

ZHANG, J., WANG, P. and QU, B. (2012). "Bank risk taking, efficiency, and law enforcement: evidence from Chinese city commercial banks", *China Economic Review*, Vol. 23, No. 2, pp. 284-295.